CPT202 Software Engineering Group Project

Module Context

This group project is a Year 3 Software Engineering Group Project. Each team will design and implement a web-based information system based on one of the project options listed in this document.

The project is intended to assess your ability to apply software engineering methods in a realistic team setting, including but not limited: requirements analysis, system design, implementation, testing, teamwork and project management, etc.

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1. General Requirements (Applicable to All Project Options)

Your system must be a web-based system and should include the following as a minimum:

1. **User authentication (register/login/logout)**
2. **Role-based access control (different users have different permissions)**
3. **Database-backed data storage**
4. **CRUD operations for core business entities**
5. **A clear business workflow with status changes**
6. **Search and/or filter functions**
7. **Usable interface for end users**
8. **Testing evidence (test cases and/or demo scenarios)**

Scope Control

- You are expected to complete a working core system first.
- Advanced features are optional and should only be attempted after the core requirements are stable.

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2. Common Deliverables (All Teams)

Unless otherwise specified by the module leader, each team should submit:

1. Software Requirements Analysis (Assignment 1)
2. Group Report and Presentation (Assignment 2)
3. Final Individual Report (Assignment 3)

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3. Project Options

Option A: Community Heritage Resource Sharing and Curation Platform

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Project Overview

Develop a web-based community heritage resource sharing and curation platform for submitting, reviewing, publishing, and archiving curated resource entries. The platform focuses on community heritage and local culture topics (e.g., places, traditions, stories, objects, or educational materials).

Contributors submit resources (e.g., images, etc) for review, administrators/reviewers moderate them before publication, and registered users can browse, search, and view approved entries.

Main User Roles

- Administrator / Reviewer
- Contributor
- Registered Viewer

Core Functional Requirements

Your system should support:

A. User and Access Management

- User registration/login/logout
- Profile maintenance
- Contributor approval by administrator

B. Resource Submission and Management

- Contributors create/edit/submit resources with metadata: title, category/topic, place, description, tags/keywords, file upload and/or external link, copyright/usage declaration, etc.
- Contributors can revise and resubmit rejected resources

C. Review and Publication Workflow

- Resource status management (minimum): Draft, Pending Review, Approved, Rejected, Archived
- Reviewer approval/rejection with feedback/comments

D. Discovery and Use

- Browse/search/filter approved resources
- View resource details
- Basic commenting/feedback on approved resources

E. Administration and Reporting

- Manage categories/tags (master data)
- Archive/unpublish resources

Business Rules (Minimum)

- Only approved contributors may submit resources
- Only approved resources are visible to general users
- Archived resources are hidden from general users

Option B: Specialist Consultation Booking System

(Supervisor: Soon Phei Tin (Soon.Tin@xjtlu.edu.cn)

TA & Groups

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Project Overview

Develop a web-based booking system for a consultancy service provider. Customers can browse specialists, check available slots, and create bookings. Staff manage specialists, time slots, bookings, and pricing.

The system should support the workflow: **booking → confirmation → cancellation/rescheduling → completion**.

Main User Roles

- Administrator / Operations Manager
- Specialist
- Customer

Core Functional Requirements

Your system should support:

A. User and Access Management

- User registration/login/logout
- Profile maintenance

B. Specialist and Availability Management

- Manage specialist records (name, expertise, level, fee, status)
- Maintain available consultation slots
- Browse/search/filter specialists by expertise/level/availability

C. Booking Workflow

- Customers book appointments by selecting specialist, date/time slot, topic/notes
- Conflict detection (prevent double booking)
- Booking status management (minimum): Pending, Confirmed, Cancelled, Completed
- Confirmation/rejection of bookings
- Cancellation/rescheduling (subject to rules)
- Automatic charge calculation (simple pricing rules)

D. Specialist and Customer Views

- Specialist schedule view
- Specialist marks appointments as completed
- Customer booking history and status tracking

E. Administration and Reporting

- Manage expertise categories/master data

Business Rules (Minimum)

- A slot cannot be booked by more than one customer
- Only confirmed bookings are valid appointments
- Completed bookings cannot be modified by customers
- Pricing must follow defined rules consistently

Option C: An online project selection system for teachers to approve

(Supervisor: Nanlin Jin (Nanlin.Jin@xjtlu.edu.cn)

TA & Groups
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Project Overview

Develop a web-based system for a university department to manage student project topic selection and agreement with teachers/staff supervisors.

Teachers publish project topics, students browse and apply, and teachers/staff make decisions. The system should support controlled allocation and agreement tracking.

Main User Roles

- Administrator
- Teacher / Staff Supervisor
- Student

Core Functional Requirements

Your system should support:

A. User and Access Management

- User registration/login/logout
- Profile maintenance

B. Project Topic Management

- Teachers/staff create and maintain project topics, including title, description, required skills, keywords/topic area, maximum number of students (if applicable), status
- Teachers/staff can publish/close/archive topics
- Admin manages project categories/tags (master data)

C. Student Search and Request Workflow

- Students browse/search/filter projects
- Students submit project requests/applications with notes
- Request status management (minimum): Pending, Accepted, Rejected, Withdrawn
- Project status management (minimum): Available, Requested, Agreed, Closed / Archived
- Teachers/staff review and accept/reject requests
- Students view request history and status

D. Allocation Control and Administration

- Rule enforcement: a student cannot hold more than one active agreed project
- Admin manages users and records

Business Rules (Minimum)

- One student may have only one active agreed project at a time
- Closed/archived projects cannot receive new requests
- Teachers/staff may manage only their own project topics
- A clear policy must be implemented for conflicting requests after acceptance

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4. Recommended Development Approach (Team Process)

You are expected to demonstrate software engineering process evidence.
Your team is encouraged to include:

- requirements analysis (use cases / user stories)
- system design (ERD, architecture, diagrams)
- sprint/task planning
- version control practice
- testing plan and evidence
- team reflection and risk management

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5. Minimum vs Advanced Scope Guidance

Minimum (Expected for all teams)

A complete core system should include:

- authentication + role-based access
- core CRUD
- workflow/status changes
- search/filter
- basic reports
- validation + testing evidence